```
$ curl -i https://the-future.org/api
HTTP/1.1 402 Payment Required
```

Pay-per-request APIs with x402 + stablecoins

UCD Workshop v2 · < 2 hours · zero real money

presented by Kostas Apostolopoulos

// testnet only. no fiat was harmed in the making of this workshop.

// boot sequence

[1] github.com/kgapos/ucd-workshop-2

[2] codespaces.new/kgapos/ucd-workshop-2 > Create Codespace

[3] Yes, I trust the authors - mostly harmless 👍

[4] Terminal › New Terminal

// it takes a couple of minutes. talk to you meanwhile.

// $ whoami

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now Principal DevOps Engineer @plasma.org



// previously, at UCD



2025 · AI agents on the blockchain
https://github.com/ucd-workshop

agents could read the chain

agents could talk about it

agents could not pay for anything

v2: give them a wallet and x402

HTTP/1.1, 1997

402

Payment Required

status: “reserved for future use”. for 29 years 🫣

// buying one API call

legacy.exe

~~sign up~~

~~verify email~~

~~credit card~~

~~API key~~

~~$29/month plan~~

x402

GET /fortune

← 402 + price

sign + retry

← 200 OK

a key pair. that’s it.

// one paid request, traced

hop	payload
client → server	GET /fortune
server → client	402 + PAYMENT-REQUIRED
client	signs off-chain. free.
client → server	GET /fortune + PAYMENT-SIGNATURE
server → facilitator	verify, then settle
facilitator → chain	USDC transfer, pays gas, ~2s
server → client	200 OK + PAYMENT-RESPONSE

// the facilitator

trusted for liveness.
not with your money.

can

- submit your signed transfer
- pay the gas
- refuse to settle

cannot

- change the amount
- change the recipient
- reuse it after 5 minutes

EIP-3009 transferWithAuthorization: exactly 0.001 USDC, to this address, for 5 minutes.

// why machines want stablecoins

`$0.001 == $0.001`	a stable price tag
`1000`	base units = a tenth of a cent
`~2s`	final. no chargebacks.
`0 accounts`	a key pair is enough
`if (worth_it) pay()`	money is just a signed message

// trade-offs: public ledger · lose the key, lose the funds · issuers can freeze

// playground: Base Sepolia

84532

chain id, an Ethereum L2 testnet

2s

per block

3009

EIP-3009 USDC from Circle

$0

real value. break things freely.

every move is public: sepolia.basescan.org

// you are mine for only 5 more slides

// the plan

0 open the Codespace

1 mint a wallet, get funded

2 put a price on your API

3 write a client that pays

4 buy from your classmates

5 stretch goals, challenges

AI tools welcome. they love the old v1 packages, so point them at @x402/*.

// the leaderboard

ucd.kgapos.com

every API. every payment. live.

// part 1: make a wallet

```
$ npm run wallet
$ npm run register -- https://ucd.kgapos.com
$ npm run balance
```

wait for 1 USDC + 0.0005 ETH. the instructor is the faucet.

// testnet key only. never put a real key in .env

// side quests: 14, easy → expert

EASY

find your payment on the explorer. who paid the gas?

MEDIUM

dynamic pricing: /bulk?count=N

HARD

an agent that shops the whole board on a budget

EXPERT

run your own facilitator. what could a rogue one do?

full list: README › Challenges

“Cypherpunks write code.”

Eric Hughes, A Cypherpunk’s Manifesto, 1993

> docs.x402.org

> github.com/x402-foundation/x402

> eips.ethereum.org/EIPS/eip-3009

> sepolia.basescan.org

// find me

web	kgapos.com
github	github.com/kgapos
x	x.com/kgapos
linkedin	linkedin.com/in/kgapos

// career opportunities for cracked engineers: plasma.org/careers

$ exit 0 // thank you!